Autonomy of Unmanned Aerial Vehicles

Practice 06 - Part A

by

Dr. Carlos Sotelo - Dr. David Sotelo

Tecnológico de Monterrey

Sep 2nd, 2024

Academic Minor

Autonomy of Unmanned Aerial Vehicles

It is an advanced level course in control engineering, in which the student implements their own control algorithms, applying the competencies developed in the focus stage of their study plan.

- It requires prior knowledge of modeling and automation and control system design.
- The student supports his proposal by the analysis of the performance of the control system.

QDrone 2

Qube-Servo 3

Aero 2

Tecnológico de Monterrey

Academic Minor

Autonomy of Unmanned Aerial Vehicles

It is an advanced level course in control engineering, in which the student implements their own control algorithms, applying the competencies developed in the focus stage of their study plan.

- It requires prior knowledge of modeling and automation and control system design.
- The student supports his proposal by the analysis of the performance of the control system.

QDrone 2

Qube-Servo 3

Aero 2

Tecnológico de Monterrey

Academic Minor

Autonomy of Unmanned Aerial Vehicles

It is an advanced level course in control engineering, in which the student implements their own control algorithms, applying the competencies developed in the focus stage of their study plan.

- It requires prior knowledge of modeling and automation and control system design.
- The student supports his proposal by the analysis of the performance of the control system.

Configuration 1

Qube-Servo 3

Configuration 2

Tecnológico de Monterrey

Academic Minor

Autonomy of Unmanned Aerial Vehicles

Qube-Servo 3

It is supplied with two standard items:

- Inverted pendulum

  The inverted pendulum is mounted to the equipment with magnets. Here, the encoder provide an additional sensor input to the system [6].

- Inertia disk

  A small aluminum part is mounted to the equipment using magnets.

Fig. 1 Inverted pendulum.

Fig. 2 Inertia disk.

Tecnológico de Monterrey

Practice 06

Context

Abstract

The PID is a well-know controller, and it is composed by three terms. The proportional term is based on the present error, the integral term depends on past errors, and the derivative term is a prediction of future errors. However, its implementation may not lead to a good system response for real-world systems. The main reason is that measured signals often includes measurement noise. Differentiating a noisy measured signal (i.e. in the derivative control) can produce a control signal with high-frequency components that yields an undesirable response and, over time, may even damage the actuator (e.g. DC motor). For this reason, in this lab practice, a new control strategy, called PV control, is proposed to regulate the angular position of the QUBE-Servo 3. Hence, analysis of the control signal is performed and compared with the PD controller.

Topics Covered

- Servo position control
- Proportional-integral-derivative control
- Design control according to specifications

Prerequisites

- Hardware Interfacing laboratory experiment
- Matlab® script of 2-D line plot

Tecnológico de Monterrey

Practice 06

Introduction

The transfer function $P(s)$ that relates the angular position and the voltage of the system, $\Theta_m(s)/V_m(s)$ obtained in Lab Practice 3 is the following,

$$P(s) = \frac{k_t}{J_{eq}R_m s^2 + k_t(k_m + \mu)s} \tag{1.1}$$

Thus, a proportional, integral, and derivative (PID) control can be implemented to regulate the angular position of the QUBE-Servo 3. Here, the control law can be mathematically expressed as

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt} \tag{1.2}$$

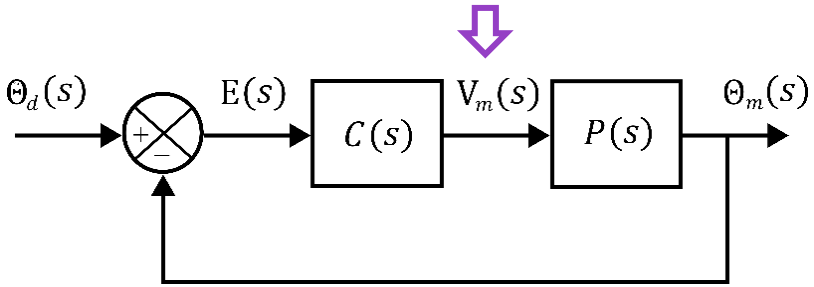


Fig. 1.1 Feedback control loop.

Practice 06

Introduction

Henceforth, based on Eq. 1.2 and Fig. 1.1, the block diagram of the feedback loop under the PID control strategy is presented in Fig. 1.2

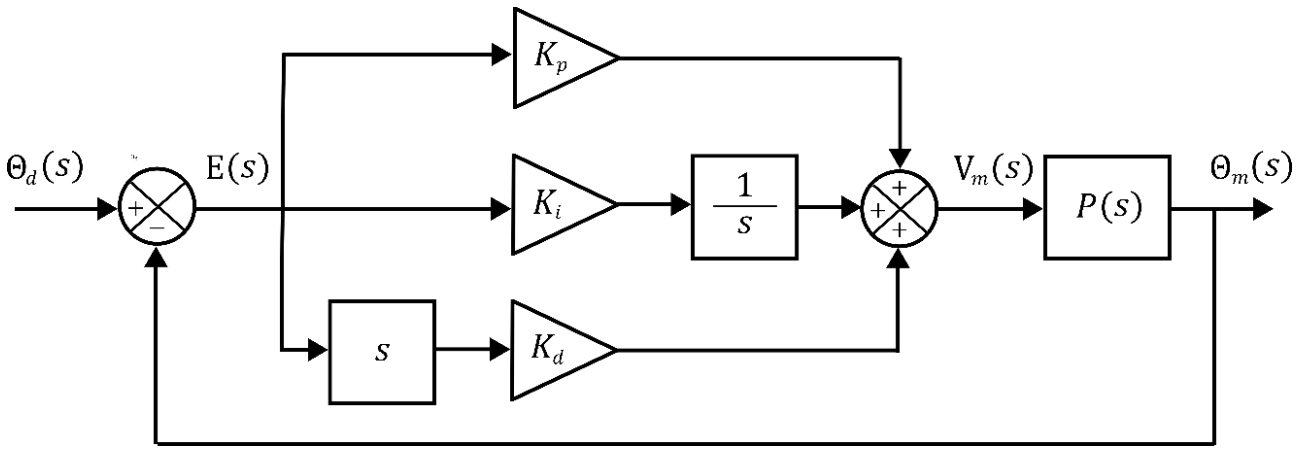


Fig. 1.2 Block diagram of PID control.

Remark 1: The PID controller may not lead to a good system response for real-world systems.

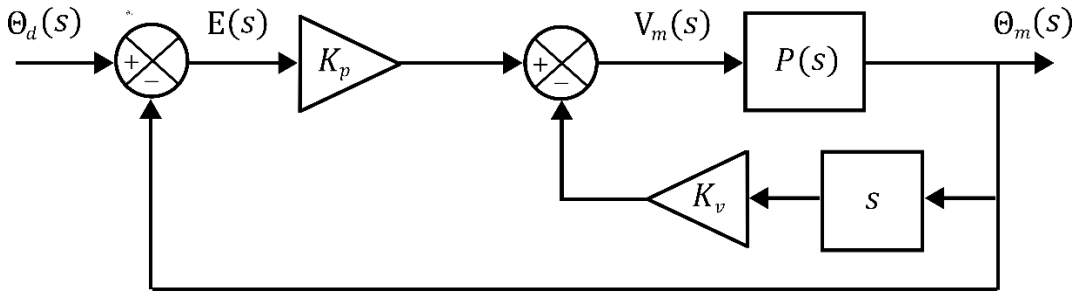
Practice 06

Development

Example: DC motor

Fig. 2 Inertia disk.

1 Obtain the mathematical expression for the control law.



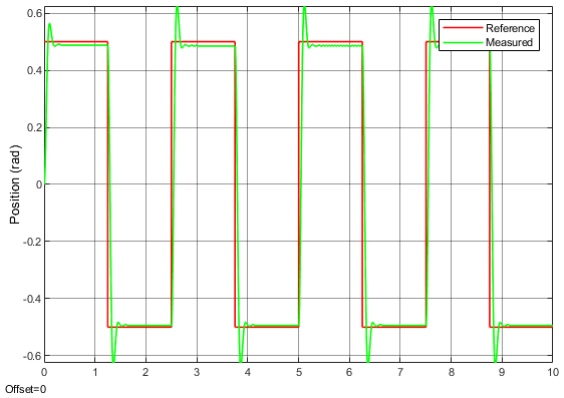
Practice 06

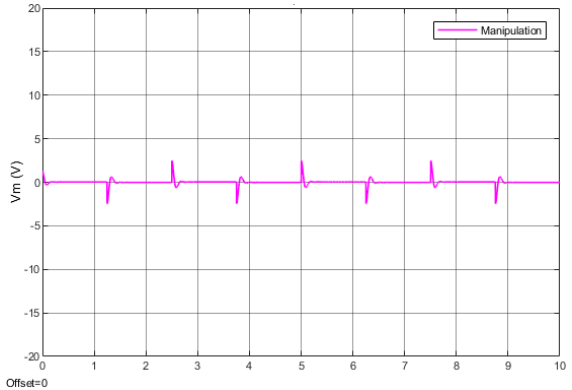
Development

Example: DC motor

Fig. 2 Inertia disk.

1. Obtain the mathematical expression for the control law.
2. Design a model of the proportional plus rate (PV) feedback control.
3. Plot results under control ($K_p = 2.5[\mathrm{V/rad}]$, $K_v = 0.05[\mathrm{V-s/rad}]$).





Tecnológico de Monterrey

Practice 06

Development

Example: DC motor

Fig. 2 Inertia disk.

1. Obtain the mathematical expression for the control law.
2. Design a model of the proportional plus rate (PV) feedback control.
3. Plot results under control ($K_p = 2.5[\mathrm{V/rad}]$, $K_v = 0.05[\mathrm{V-s/rad}]$).
4. Perform a qualitative analysis of the QUARC® controller.
   - **i.** Vary $K_p = 1 - 4\ [\mathrm{V/rad}]$ keep $K_v = 0\ [\mathrm{V-s/rad}]$
   - **ii.** Keep $K_p = 2.5\ [\mathrm{V/rad}]$ vary $K_v = 0 - 0.15\ [\mathrm{V-s/rad}]$

Tecnológico de Monterrey

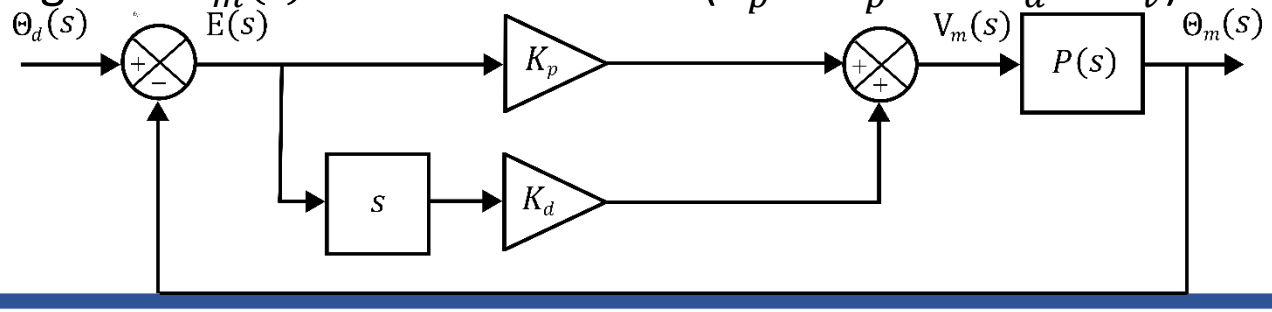
Practice 06

Development

Example: DC motor

Fig. 2 Inertia disk.

1. Obtain the mathematical expression for the control law.
2. Design a model of the proportional plus rate (PV) feedback control.
3. Plot results under control ($K_p = 2.5[\mathrm{V/rad}]$, $K_v = 0.05[\mathrm{V-s/rad}]$).
4. Perform a qualitative analysis of the QUARC® controller.
5. Obtain the closed-loop transfer function $\Theta_m(s)/\Theta_d(s)$.
6. Find K_p and K_v to achieve $\%MP = 2.5\ \%$ and $t_p = 0.15$ [sec] in $\theta_m(t)$
7. Regulate $\theta_m(t)$ with a PD control ($K_p = K_p$ and $K_d = K_v$).



Tecnológico de Monterrey

Practice 06

Development

Example: DC motor

Fig. 2 Inertia disk.

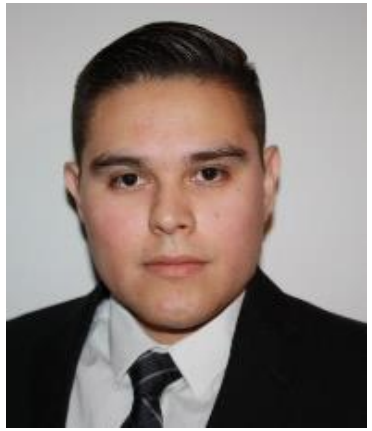
1. Obtain the mathematical expression for the control law.
2. Design a model of the proportional plus rate (PV) feedback control.
3. Plot results under control ($K_p = 2.5[\mathrm{V/rad}]$, $K_v = 0.05[\mathrm{V-s/rad}]$).
4. Perform a qualitative analysis of the QUARC® controller.
5. Obtain the closed-loop transfer function $\Theta_m(s)/\Theta_d(s)$.
6. Find K_p and K_v to achieve $\%MP = 2.5\ \%$ and $t_p = 0.15\ [\mathrm{sec}]$ in $\theta_m(t)$
7. Regulate $\theta_m(t)$ with a PD control ($K_p = K_p$ and $K_d = K_v$).
8. Using PV control, modify Simulink model to set the reference with a second QUBE-Servo 3.
9. Conclusions.

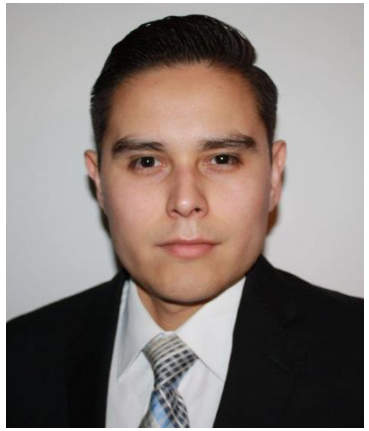
Tecnológico de Monterrey

Thank you!

Professor
Carlos Sotelo
carlos.sotelo@tec.mx

Professor
David Sotelo
david.sotelo@tec.mx





Tecnológico de Monterrey

Sep 2nd, 2024